

PhD Resume Sample

Mechanical Engineer

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SUMMARY

Extensive experience with applying analytical and numerical methods (such as the finite element method) to model a broad range of systems from molecular structures to large-scale mechanical structures. Proven track record of creating and improving new computational methods to perform dynamic and static analysis of otherwise intractable engineering and biological systems. Strong ability to collaborate and work in a team environment on multi-disciplinary projects. Legally authorized to work in the United States (Green Card holder).

EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA 2011

Ph.D., Department of Mechanical Engineering.

- Thesis: "Contributions to the analysis of proteins" under the supervision of Prof. Jones and Prof. Smith
- GPA: 5.0/5.0 (Awarded an A+ grade for all courses. Only one or two people in each course get A+.)

Sharif University of Technology, Tehran, IRAN 2005

M.Sc., Department of Mechanical Engineering.

- Thesis: "Online control of needle injection into soft tissue using the finite element method"
- GPA: 18.62/20.0 (Ranked in top 5%)

University of Tehran, Tehran, IRAN 2003

B.Sc., Department of Mechanical Engineering.

- GPA: 17.68/20.0 (Class Rank: 2)

SKILLS

- **Computer:** Commercial finite element software programs: ADINA (founded and owned by my Ph.D. and postdoctoral advisor, Prof. KJ Bathe), ABAQUS, ANSYS; MeshLab (a mesh processing program); MATLAB; Fortran; AutoCAD; molecular viewers: PyMOL, VMD, UCSF Chimera; CHARMM (a molecular dynamics program); Adobe Illustrator.
- **Analytical:** Finite element method; optimization; stochastic simulation: Langevin and Brownian dynamics simulation; statistical analysis; multi-scale modeling; atomistic modeling; continuum modeling; bioinformatics; biomechanics; computational biology; molecular biology; biophysics; solid mechanics; fluid mechanics; controls.
- **Language:** English (fluent); Persian (native); Arabic (basic).

EXPERIENCE

Department of Mechanical Engineering, MIT, Cambridge, MA, USA Oct. 2011–current

Postdoctoral Associate

- Led project team that developed a coarse-grained finite element framework for the Brownian dynamics of macromolecular proteins that are inaccessible to available molecular dynamics algorithms.
- Created a model to calculate the diffusion coefficients and Brownian dynamics of DNA origami structures as part of a project in collaboration with researchers from MIT, Harvard, University of Michigan, Arizona State University, and Max Planck Institute. No other models are currently available.
- Member of team that developed a coarse-grained three-dimensional hydrodynamic model of semi-flexible filaments that resulted in several orders-of-magnitude reduction in computational cost.
- Collaborated with other engineers to improve a well-known implicit time-integration scheme that is widely used in engineering problems and in numerous commercial software tools. The improved version of the scheme has already been implemented in ADINA.

Department of Mechanical Engineering, MIT, Cambridge, MA, USA Jan. 2007–Jun. 2011

Research Assistant

- Improved a widely used eigenvalue solver to substantially reduce the computational cost of calculating the eigen-solutions of large-scale engineering and bioengineering systems. The improved version of the eigenvalue solver is currently used in ADINA.
- Made novel discoveries into the shape and function of complex proteins, the results of which have been included in comprehensive government and research databases (such as the Protein Data Bank) and utilized by leading research companies.
- Developed a coarse-grained finite element framework for the diffusion coefficients of proteins.

Department of Mechanical Engineering, MIT, Cambridge, MA, USA Fall 2007, Fall 2008, Fall 2010

Teaching Assistant, "Finite Element Analysis of Solids and Fluids I" & "Mechanics and Materials I"

- Prepared and presented lectures and recitations, supported term projects, helped students with course materials, and graded homework and